Valori și vectori proprii CURS 11

Fie $A \in \mathbb{R}^{m \times m}$ a.i. $\boxed{A \cdot x = \lambda \cdot x}$, $\lambda \in \mathbb{C}$, $x \in \mathbb{C}^m$

$\lambda(A) = \{\lambda_1, \dots, \lambda_m\}$ spectrul matricei A

$\rho(A) = \max_{\lambda_i \in \lambda(A)} |\lambda_i|$ raza spectrală a matricei A

$P_A(\lambda) = |\lambda I_m - A|$ polinom caracteristic.

Proprietăți

1) $\lambda(A - \mu I_m) = \{\lambda_1 - \mu, \lambda_2 - \mu, \dots, \lambda_m - \mu\}$; $\mu \in \mathbb{C}$
2) $\lambda(A^{-1}) = \{\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_m}\}$
3) $\lambda(A^k) = \{\lambda_1^k, \lambda_2^k, \dots, \lambda_m^k\}$
4) $\lambda\left((A - \mu I_m)^{-1}\right) = \{\frac{1}{\lambda_1 - \mu}, \frac{1}{\lambda_2 - \mu}, \dots, \frac{1}{\lambda_m - \mu}\}$; $\mu \in \mathbb{C} \setminus \lambda(A)$

Demonstrații

1) $Ax = \lambda x \Rightarrow Ax - \mu I_m x = \lambda x - \mu I_m x \Rightarrow (A - \mu I_m)x = (\lambda - \mu)x$
2) $Ax = \lambda x \Rightarrow x = \lambda A^{-1}x \Rightarrow \frac{1}{\lambda} x = A^{-1}x$
3) $Ax = \lambda x \Rightarrow A^2 x = \lambda Ax = \lambda \cdot \lambda x = \lambda^2 x \xrightarrow{\text{Inductie}} A^k x = \lambda^k x$.
4) $(A - \mu I_m)x = (\lambda - \mu)x \Rightarrow x = (\lambda - \mu)(A - \mu I_m)^{-1}x \Rightarrow$
$\Rightarrow (A - \mu I_m)^{-1}x = \frac{1}{\lambda - \mu} x$.

Metoda puterii directe

Presupunem $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_m|$

λ_1 s.m. Valoare proprie dominantă.

M.P.D. calculează (λ_1, x_1).

Fie șirul: $y^{(0)} \neq 0$

$\Downarrow$ converge către x_1

$y^{(1)} = Ay^{(0)}$ $\quad \lambda^{(1)}$

$y^{(2)} = A \cdot y^{(1)} = A \cdot A \cdot y^{(0)} = A^2 y^{(0)}$ $\quad \lambda^{(2)}$

$\dots\dots$

$y^{(k)} = A \cdot y^{(k-1)} = \dots = A^k \cdot y^{(0)}$ $\quad \lambda^{(k)} = ?$

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$$A \cdot y^{(k)} = \lambda^{(k)} \cdot y^{(k)}$$

$$\left(y^{(k)}\right)^T A \cdot y^{(k)} = \lambda^{(k)} \left(y^{(k)}\right)^T \cdot y^{(k)}$$

$$\lambda^{(k)} = \frac{\left(y^{(k)}\right)^T \cdot A \cdot y^{(k)}}{\left(y^{(k)}\right)^T \cdot y^{(k)}}$$ catul Rayleigh

Algoritm OCTAVE

Di : $A, y^{(0)}$, maxiter, tol

for k = 1 : maxiter

- $z = A \cdot y^{(k-1)}$
- $y^{(k)} = z / \text{norm}(z)$
- $\lambda^{(k)} = (y^{(k)})^T A\, y^{(k)}$
- → test convergență alg.

test convergență

$\| y^{(k)} - y^{(k-1)} \| < \varepsilon$

$\| \lambda^{(k)} - \lambda^{(k-1)} \| < \varepsilon$

$\| A y^{(k)} - \lambda^{(k)} \cdot y^{(k)} \| < \varepsilon$

Exemplu

- $A = \begin{bmatrix} -7 & 2 \\ 8 & -1 \end{bmatrix}$; $y^{(0)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$; $\lambda_1 = -9$; $\lambda_2 = 1$

$$y^{(1)} = \frac{A y^{(0)}}{\| A y^{(0)} \|} = \begin{bmatrix} -0,65 \\ 0,75 \end{bmatrix} \quad \lambda^{(1)} = \left(y^{(1)}\right)^T A\, y^{(1)} = -8,55$$

$$y^{(2)} = \frac{A y^{(1)}}{\| A y^{(1)} \|} = \begin{bmatrix} 0,71 \\ -0,71 \end{bmatrix} \quad \lambda^{(2)} = \left(y^{(2)}\right)^T A\, y^{(2)} = -9,04$$

....

- $A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$; $y^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\lambda_1 = 1$; $\lambda_2 = -1$

$$y^{(1)} = \frac{A y^{(0)}}{\| A y^{(0)} \|} = \begin{bmatrix} 0,89 \\ -0,44 \end{bmatrix} \quad \lambda^{(1)} = \left(y^{(1)}\right)^T A\, y^{(1)} = 0,2$$

$$y^{(2)} = \frac{A y^{(1)}}{\| A y^{(1)} \|} = \begin{bmatrix} 0,707 \\ 0,707 \end{bmatrix} \quad \lambda^{(2)} = \left(y^{(2)}\right)^T A\, y^{(2)} = 0,5$$

.....

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Exemplu

Convergența șirului $y^{(k)}$

$y^{(k)} = \frac{A^k y^{(0)}}{\| A^k y^{(0)} \|}$. Notăm $\gamma_k = \frac{1}{\|A^k y^{(0)}\|}$

Fie $y^{(0)} = \sum_{i=1}^{n} \alpha_i x_i$

$$y^{(k)} = \gamma_k \cdot A^k \sum_{i=1}^{n} \alpha_i x_i = \gamma_k \sum_{i=1}^{n} \alpha_i A^k x_i = \gamma_k \sum_{i=1}^{n} \alpha_i \lambda_i^k \cdot x_i =$$

$$= \gamma_k \alpha_1 \lambda_1^k x_1 + \gamma_k \sum_{i=2}^{n} \alpha_i \lambda_i^k x_i =$$

$$= \gamma_k \alpha_1 \lambda_1^k \left[x_1 + \sum_{i=1}^{n} \frac{\alpha_i}{\alpha_1} \left(\frac{\lambda_i}{\lambda_1} \right)^k x_i \right]$$

Calculare cea mai mică val proprie

Pp. $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_{m-1}| > |\lambda_m|$

$\lambda(A^{-1}) = \left\{ \frac{1}{\lambda_1}; \frac{1}{\lambda_2}; \dots; \frac{1}{\lambda_m} \right\}$

$\left| \frac{1}{\lambda_m} \right| > \left| \frac{1}{\lambda_{m-1}} \right| \geq \dots \geq \left| \frac{1}{\lambda_2} \right| > \left| \frac{1}{\lambda_1} \right|$

Algoritm

Di: $A, y^{(0)}$, maxiter, tol

for $k = 1 : \text{maxiter}$

$z = A^{-1} y^{(k-1)}$

$y^{(k)} = z / \text{norm}(z)$

$\lambda^{(k)} = (y^{(k)})^T A^{-1} y^{(k)}$

$\| A^{-1} y^{(k)} - \lambda^{(k)} y^{(k)} \| < \text{tol}$ (test convergență)

return $1 / \lambda^{(k)}$

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Metoda puterii inverse

Di : $A, y^{(0)}$, maxiter, tol, μ

for $k = 1 : maxiter$

$z = (A - \mu I)^{-1} y^{(k-1)}$

$y^{(k)} = z / norm(z)$

$\lambda^{(k)} = (y^{(k)})^T A y^{(k)}$

$\| Ay^{(k)} - \lambda^{(k)} y^{(k)} \| < tol$ (test convergență)

Localizarea valorilor proprii

Teoremă: Fie $A \in \mathbb{R}^{n \times n}$ și cercurile Gershgorin definite astfel:

$$D_i = \{ z \in \mathbb{C} \mid |z - a_{ii}| \leq \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}| \}, \quad i = 1 : n$$

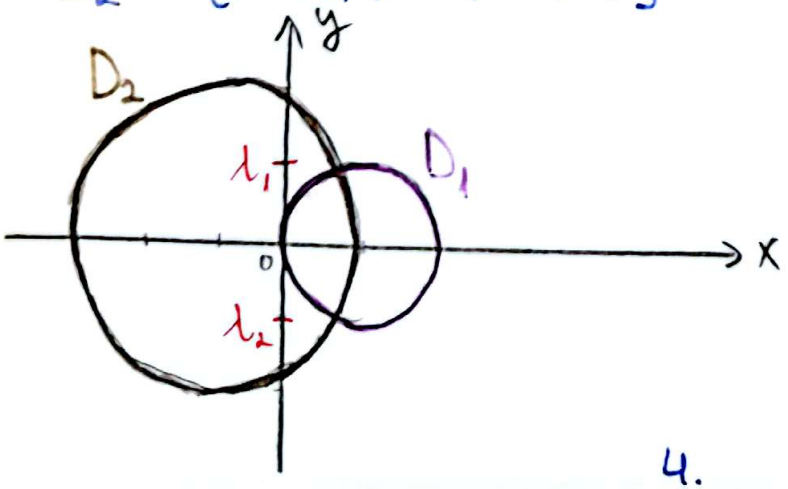
Atunci $\lambda(A) \subset \bigcup_{i=1}^{n} D_i$.

- centrul cerc D_i : $(a_{ii}, 0)$
- raza cerc D_i : $\sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|$

Exemplu: $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$; $\lambda_1 = i$; $\lambda_2 = -i$

$D_1 = \{ z \in \mathbb{C} \mid |z - 1| \leq |-1| \}$

$D_2 = \{ z \in \mathbb{C} \mid |z + 1| \leq |2| \}$



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Exemplu

Descompunerea valorilor singulare (SVD) CURS 12-13

Fie $A \in \mathbb{R}^{m \times n}$. $A = U \cdot \Sigma \cdot V^T$, unde

- $U = [u_1, \dots, u_m]$ matrice ortogonala, $U \in \mathbb{R}^{m \times m}$; $u_1, \dots, u_m$ vectorii singulari stanga ai matricei A.
- $V^T = \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix}$ matrice ortogonală, $V^T \in \mathbb{R}^{n \times n}$; $v_1, \dots, v_n$ vectorii singulari dreapta ai matricii A.
- $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_p)$; $p = \min(m,n)$, $\Sigma \in \mathbb{R}^{m \times n}$; $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$ valorile singulare ale lui A.

Calculare valori singulare

$\boxed{\sigma_i = \sqrt{\lambda_i}}$ unde $\lambda_i \in \lambda(A^TA)$, $i = 1:p$.

Demonstratie: $A = U \Sigma V^T \Rightarrow AV = U \Sigma \underbrace{V^T V}_{I} = U\Sigma \Rightarrow$

$\Rightarrow [Av_1\ Av_2 \dots Av_n] = [\sigma_1 u_1\ \sigma_2 u_2 \dots \sigma_n u_n] \Rightarrow$

$\Rightarrow Av_i = \sigma_i u_i$ (1)

$A = U \Sigma V^T \Rightarrow A^T = (U \Sigma V^T)^T = V \Sigma U^T \Rightarrow A^T U = V \Sigma \Rightarrow$

$\Rightarrow [A^T u_1\ A^T u_2 \dots A^T u_n] = [\sigma_1 v_1\ \sigma_2 v_2 \dots \sigma_n v_n] \Rightarrow$

$\Rightarrow A^T u_i = \sigma_i v_i$ (2)

$\text{din (1)} \Rightarrow A^TA v_i = \sigma_i A^T u_i \overset{(2)}{=} \sigma_i \sigma_i v_i \Rightarrow A^TA v_i = \sigma_i^2 v_i$ (3)

$\text{din (2)} \Rightarrow AA^T u_i = \sigma_i A v_i \overset{(1)}{=} \sigma_i \sigma_i u_i \Rightarrow AA^T u_i = \sigma_i^2 u_i$ (4)

Calculare V

$V = [v_1 \dots v_n] \in \mathbb{R}^{n \times n}$

$\text{din (3)} \Rightarrow A^TA v_i = \sigma_i^2 v_i \Rightarrow v_i$ sunt vectorii proprii ai matricii A^TA

MPi

Calculare U

$U = [u_1 \dots u_m] \in \mathbb{R}^{m \times m}$

$\text{din (1)} \Rightarrow Av_i = \sigma_i u_i \Rightarrow \boxed{u_i = \frac{1}{\sigma_i} Av_i,\ i = 1:r}$

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$U = [u_1 \dots u_r \; u_{r+1} \dots u_m]$ (Vectori $u_{r+1}, \dots, u_m$ se calculează folosind Gram-Schmidt)

Exemplu $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \Rightarrow A^TA = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 14 & 20 \\ 20 & 29 \end{bmatrix}$

$\lambda(A^TA) = \{42,86 ; 0,14\} \Rightarrow \cdot \Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix}$

($\lambda_1 = 42,86$, $\lambda_2 = 0,14$)

$\sigma_1 = \sqrt{\lambda_1} = 6,54$; $\sigma_2 = \sqrt{\lambda_2} = 0,37$

- $V = [v_1 \; v_2] \in \mathbb{R}^{2\times 2}$

$A^TAv_1 = \sigma_1^2 v_1 \Rightarrow v_1 = \begin{bmatrix} 0,56 \\ 0,82 \end{bmatrix}$ MPi...

$A^TAv_2 = \sigma_2^2 v_2 \Rightarrow v_2 = \begin{bmatrix} -0,82 \\ 0,56 \end{bmatrix}$

- $U = [u_1 \; u_2 \; u_3] \in \mathbb{R}^{3\times 3}$

$u_1 = \frac{1}{\sigma_1} A v_1 = [0,33 \;\; 0,55 \;\; 0,76]^t$

$u_2 = \frac{1}{\sigma_2} A v_2 = [0,84 \;\; 0,17 \;\; -0,5]^t$

u_3 se calc cu Gram-Schmidt.

Proprietăți SVD, $A \in \mathbb{R}^{m\times m}$

1) $\|A\|_2 = \sigma_1$

$\|A\|_2 = \|U\Sigma V^T\|_2 = \|\Sigma V^T\|_2 = \|\Sigma\|_2 = \sqrt{\rho(\Sigma\Sigma^T)} =$

$= \sqrt{\max\{\sigma_1^2, \sigma_2^2, \dots, \sigma_m^2\}} = \sqrt{\sigma_1^2} = \sigma_1$

2) $\|A\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_m^2}$

$\|A\|_F = \|U\Sigma V^T\|_F = \|\Sigma\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_m^2}$

3) $\|A^{-1}\|_2 = \frac{1}{\sigma_m}$

4) $\text{cond}(A) \overset{\text{def}}{=} \|A\|_2 \cdot \|A^{-1}\|_2 = \frac{\sigma_1}{\sigma_m}$

5) $\text{rang}(A) =$ nr de val singulare > 0.

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Matrici asemenea

CURS 14

$A, B \in \mathbb{R}^{n \times n}$ sunt asemenea dacă $\exists T$ nesingulară a.î. $B = T^{-1}AT$.

Proprietăți

1) $\boxed{\lambda(A) = \lambda(B)}$

Demonstrație: $\det(B - \lambda I) = \det(T^{-1}AT - \lambda T^{-1}IT) =$

$= \det(T^{-1}(A - \lambda I)T) = \det(T^{-1}) \cdot \det(A - \lambda I) \cdot \det T =$

$= \det(A - \lambda I)$

2) Fie (λ, x) pereche proprie pt A : $Ax = \lambda x$

(λ, y) pereche proprie pt B : $By = \lambda y$

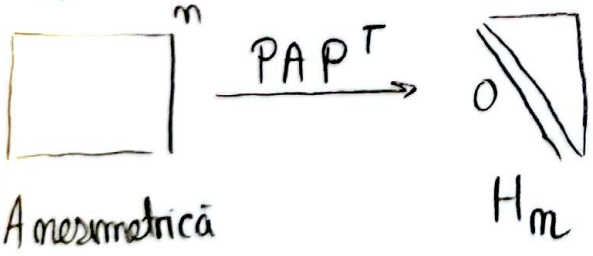
$B(T^{-1}x) = T^{-1}ATT^{-1}x = T^{-1}Ax = T^{-1}\lambda x = \lambda T^{-1}x \Rightarrow$

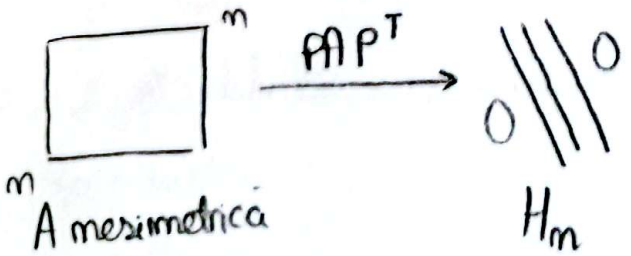
$\Rightarrow B(T^{-1}x) = \lambda(T^{-1}x) \Rightarrow y = T^{-1}x \Rightarrow \boxed{x = Ty}$

Aducerea la forma superior Hessenberg

Fie $A \in \mathbb{R}^{n \times n}$. Orice matrice A poate fi adusă la forma superior Hessenberg H_n folosind transformarea de asemănare ortogonală.

$H_n = P \cdot A \cdot P^T$, unde P matrice ortogonală.





tridiagonală simetrică
(sup Hessenberg + simetrică)

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$P = P_{n-2} \cdot \ldots \cdot P_2 P_1$; $P_1, \ldots, P_{n-2}$ matrici ortogonale Householder.

$PAP^T = P_{n-2} \ldots P_2 P_1 A P_1^T P_2^T \ldots P_{n-2}^T$

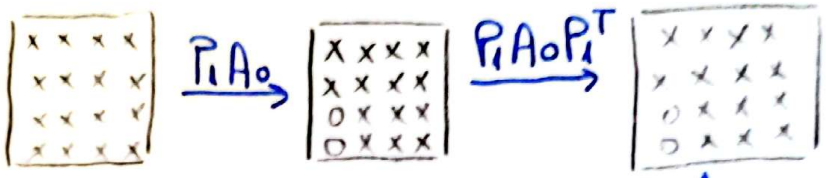
Notăm $A_0 = A \Rightarrow A_1 = P_1 A_0 P_1^T$

$A_2 = P_2 A_1 P_2^T$

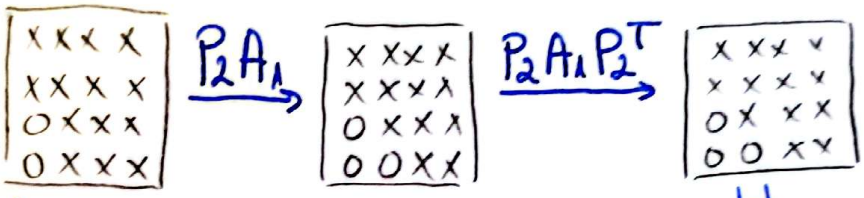
$\ldots\ldots\ldots$

$H_n = P_{n-2} A_{n-3} P_{n-2}^T$

Dacă $n=4$:



A_0 nesimetrică $\qquad A_1$



$A_1 \qquad H_n$

Pas 1: $P_1 = \begin{bmatrix} i_1 & 0 \\ 0 & \hat{P}_1 \end{bmatrix}$

$\hat{P}_1 = ?$ a.î. $\hat{P}_1 \begin{bmatrix} a_{21} \\ a_{31} \\ \vdots \\ a_{n1} \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

$\hat{P}_1 = I_{n-1} - 2\frac{v_1 v_1^T}{v_1^T v_1}$, $\hat{P}_1$ matrice Householder $(n-1)\times(n-1)$

$P_1 A_0 P_1^T = \begin{bmatrix} x & x & \ldots & x \\ x & x & \ldots & x \\ 0 & a_{32} & \ldots & x \\ \vdots & \vdots & \ldots & \vdots \\ 0 & a_{n2} & \ldots & x \end{bmatrix} = A_1$

Pas 2 $P_2 = \begin{bmatrix} I_2 & 0 \\ 0 & \hat{P}_2 \end{bmatrix}$

$\hat{P}_2 = ?$ a.î. $\hat{P}_2 \begin{bmatrix} a_{32} \\ a_{42} \\ \vdots \\ a_{n2} \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

$\hat{P}_2 = I_{n-2} - 2\frac{v_2 v_2^T}{v_2^T v_2}$, $\hat{P}_2$ matrice Householder $(n-2)\times(n-2)$

$A_2 = P_2 A_1 P_2^T \ldots\ldots$

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Exemplu

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 1 & 0 \\ 0 & \hat{P}_1 \end{bmatrix} ; A_0 = A$$

$\hat{P}_1$ a.s. $\hat{P}_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$

$$\hat{P}_1 = I_2 - 2\frac{v_1 \cdot v_1^T}{v_1^T v_1}$$

$$v_1 = \begin{bmatrix} 1+\sigma_1 \\ 1 \end{bmatrix} ; \sigma_1 = \text{sign}(1) \cdot \left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{2}$$

$$\Rightarrow \hat{P}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{2}{\sqrt{4+2\sqrt{2}}} \begin{bmatrix} 1+\sqrt{2} \\ 1 \end{bmatrix} \begin{bmatrix} 1+\sqrt{2} & 1 \end{bmatrix} = \begin{bmatrix} -0,7 & -0,7 \\ -0,7 & 0,7 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0,7 & -0,7 \\ 0 & -0,7 & 0,7 \end{bmatrix} \Rightarrow A_1 = P_1 A_0 P_1^T = \begin{bmatrix} 0 & -2,12 & 0,7 \\ -1,41 & 3,5 & -0,5 \\ 0 & 1,5 & -0,5 \end{bmatrix}$$

Algoritmul QR

Forma Schur reală a unei matrice

$A \in \mathbb{R}^{n \times n}$. Există matricea ortogonală $Q \in \mathbb{R}^{n \times n}$ a.î. $S = Q^T A Q$, unde S este o matrice cvasi superior triunghiulară

$$S = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1n} \\ 0 & s_{22} & \dots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_{nn} \end{bmatrix}$$

Algoritmul QR fără deplasare pt A matrice nesimetrică

$A_0 = H_n$

$A_0 = Q_0 R_0$

$A_1 = R_0 Q_0 = Q_1 R_1$

$A_2 = R_1 Q_1 = Q_2 R_2$

$\dots\dots\dots\dots$

$A_k = R_{k-1} Q_{k-1}$

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Arăt că A_1 este ortogonal asemenea cu A_0

$$\left.\begin{aligned} A_1 &= R_0 Q_0 \\ A_0 &= Q_0 R_0 \Rightarrow R_0 = Q_0^T A_0 \end{aligned}\right\} \Rightarrow A_1 = Q_0^T A_0 Q_0$$

Arăt că A_2 este ortogonal asemenea cu A_0

$$\left.\begin{aligned} A_2 &= R_1 Q_1 \\ A_1 &= Q_1 R_1 \Rightarrow R_1 = Q_1^T A_1 \end{aligned}\right\} \Rightarrow A_2 = Q_1^T A_1 Q_1 =$$
$$= Q_1^T Q_0^T A_0 Q_0 Q_1 =$$
$$= (Q_0 Q_1)^T A Q_0 Q_1$$

Algoritm

calculăm H_n

$A_0 = H_n$

for $k = 0 : maxiter$

$[Q_k \; R_k] = householder(A_k)$

$A_{k+1} = R_k \cdot Q_k$

Alg QR cu deplasare simplă pt A matrice nesimetrică

calculăm H_n

$A_0 = H_n$

for $k = 0 : maxiter$

$\mu^{(k)} = A_k(n,n)$

$[Q_k \; R_k] = householder(A_k - \mu^{(k)} I_n)$

$A_{k+1} = R_k \cdot Q_k + \mu^{(k)} I_n$

Alg QR fără deplasare pt A matrice simetrică

calculăm H_n

$A_0 = H_n$

for $k = 0 : maxiter$

$[Q_k \; R_k] = householder(A_k)$

$A_{k+1} = R_k \cdot Q_k$

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Alg QR cu deplasare simplă pt A simetrică

calculăm H_m

$A_0 = H_m$

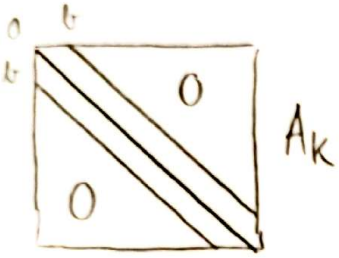
for $k = 0 : maxiter$

calculăm $\mu^{(k)}$

$[Q_k \; R_k] = householder(A_k - \mu^{(k)} I_m)$

$A_{k+1} = R_k \cdot Q_k + \mu^{(k)} \cdot I_m$

Calculare $\mu^{(k)}$



Fie $E_k = \begin{bmatrix} a_{m-1}^{(k)} & b_{m-1}^{(k)} \\ b_{m-1}^{(k)} & a_m^{(k)} \end{bmatrix}$

$$|E_k - \lambda I_2| = \begin{vmatrix} a_{m-1}^{(k)} - \lambda & b_{m-1}^{(k)} \\ b_{m-1}^{(k)} & a_m^{(k)} - \lambda \end{vmatrix} =$$

$$= (a_{m-1}^{(k)} - \lambda)(a_m^{(k)} - \lambda) - (b_{m-1}^{(k)})^2 =$$

$$= \lambda^2 - \lambda(a_m^{(k)} + a_{m-1}^{(k)}) + a_m^{(k)} \cdot a_{m-1}^{(k)} - (b_{m-1}^{(k)})^2$$

$\Rightarrow \mu^{(k)}$ = valoarea proprie a matricei E_k cea mai apropiată de elementul $A_k(m,m)$

Interpolări CURS 15 - 16

Fie $f : [x_0, x_m] \to \mathbb{R}$ cunoscută doar în punctele $x_0, \dots, x_m$ prim valorile $f(x_0), \dots, f(x_m)$. $f(x) = ?$, $x \in [x_0, x_m]$

$\{x_0, \dots, x_m\}$ s.n. suportul interpolării

$P_m(x)$ s.n polinom de interpolare de grad m

$P_m(x) = a_0 u_0(x) + a_1 u_1(x) + \dots + a_m u_m(x)$, unde $\{u_0(x), \dots, u_m(x)\}$ s.n baza interpolării

$f(x) \simeq P_m(x)$, $x \in [x_0, x_m] \setminus \{\text{suportul interpolării}\}$

Pentru a afla $a_0, \dots, a_m$ folosim condiția de interpolare:

$$f(x_i) = P_m(x_i) \; ; \; i = 0 : m$$

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Polinoame Bernstein de gradul n

$$B_{i,m}(t) = C_m^i (1-t)^{m-i} \cdot t^i, \quad i = 0:m, \quad t\text{-parametru}, \quad t \in [0,1]$$

pt $n=1$: $B_{0,1}(t) = C_1^0 (1-t)^1 \cdot t^0 = 1-t$

$B_{1,1}(t) = C_1^1 (1-t)^0 \cdot t^1 = t$

pt $n=2$: $B_{0,2}(t) = C_2^0 (1-t)^2 \cdot t^0 = (1-t)^2$

$B_{1,2}(t) = C_2^1 (1-t)^1 \cdot t^1 = 2t(1-t)$

$B_{2,2}(t) = C_2^2 (1-t)^0 \cdot t^2 = t^2$

pt $n=3$: $B_{0,3}(t) = C_3^0 (1-t)^3 \cdot t^0 = (1-t)^3$

$B_{1,3}(t) = C_3^1 (1-t)^2 \cdot t^1 = 3t(1-t)^2$

$B_{2,3}(t) = C_3^2 (1-t)^1 \cdot t^2 = 3t^2(1-t)$

$B_{3,3}(t) = C_3^3 (1-t)^0 \cdot t^3 = t^3$

.....

Interpolare polinomială

bază de interpolare : $\{1, x, x^2, \dots, x^m\}$

$P_m(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m$

condiții de interpolare : $f(x_i) = P_m(x_i)$, $i = 0:m$

$$\begin{cases} a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_m x_0^m = f(x_0) \\ a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_m x_1^m = f(x_1) \\ \dots \\ a_0 + a_1 x_m + a_2 x_m^2 + \dots + a_m x_m^m = f(x_m) \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_m & x_m^2 & \dots & x_m^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \dots \\ a_n \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ \dots \\ f(x_m) \end{bmatrix}$$

SEL cu $n+1$ necunoscute și $n+1$ ec liniare

matrice Vandermonde (rău condiționată)

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Interpolare Lagrange

bază de interpolare : $\{L_0(x), L_1(x), \dots, L_n(x)\}$ multiplicatorii Lagrange

$$L_0(x) = \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)}$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)}$$

$$L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^{n} \frac{x-x_i}{x_k-x_i}$$

polinoame de grad n

$$P_n(x) = \sum_{k=0}^{n} a_k \cdot L_k(x), \text{ unde } a_k = f(x_k)$$

polinom de interpolare Lagrange

Exemplu :

	x_0	x_1	x_2	x_3
x	1	2	3	4
$f(x)$	1	2	0	1
	a_0	a_1	a_2	a_3

$$P_3(x) = a_0 L_0(x) + a_1 L_1(x) + a_2 L_2(x) + a_3 L_3(x)$$

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-2)(x-3)(x-4)}{(-1)(-2)(-3)}$$

$$\dots\dots$$

$$L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x-1)(x-2)(x-3)}{3 \cdot 2 \cdot 1}$$

Interpolare Newton

F_0 – dif. divizate de ordin 0

F_1 – dif divizate de ordin 1

F_2 – dif divizate de ordin 2

$$F_0[x_0] = f(x_0) \longrightarrow F_1[x_0, x_1] = \frac{F_0[x_0] - F_0[x_1]}{x_0 - x_1} \longrightarrow F_2[x_0, x_1, x_2] = \frac{F_1[x_0, x_1] - F_1[x_1, x_2]}{x_0 - x_2}$$

$$F_0[x_1] = f(x_1) \longrightarrow F_1[x_1, x_2] = \frac{F_0[x_1] - F_0[x_2]}{x_1 - x_2}$$

$$F_0[x_2] = f(x_2)$$

$$\dots\dots \qquad \dots\dots$$

$$F_0[x_n] = f(x_n)$$

$$F_n[x_0, x_1, \dots, x_n] = \frac{F_{n-1}[x_0, \dots, x_{n-1}] - F_{n-1}[x_1, \dots, x_n]}{x_0 - x_n}$$

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$$f(x) = f(x_0) + (x-x_0)F_1[x_0,x_1] + (x-x_0)(x-x_1)F_2[x_0,x_1,x_2] + \dots +$$
$$+ (x-x_0)(x-x_1)\dots(x-x_{m-1})F_m[x_0,x_1,\dots,x_{m-1}] + (x-x_0)(x-x_1)\dots$$
$$(x-x_{m-1})(x-x_m)F_{m+1}[x,x_0,\dots,x_m]$$

Baza de interpolare: $\{\underbrace{1}_{u_0(x)}, \underbrace{x-x_0}_{u_1(x)}, \underbrace{(x-x_0)(x-x_1)}_{u_2(x)}, \dots, \underbrace{(x-x_0)\dots(x-x_{m-1})}_{u_m(x)}\}$

$a_0 = f(x_0) = F_0[x_0]$, $a_1 = F_1[x_0,x_1], \dots, a_m = F_m[x_0,x_1,\dots,x_m]$

Exemplu

x	x_0 1	x_1 2	x_2 3	x_3 4
$f(x)$	1	2	0	1

x	F_0	F_1	F_2
$x_0 = 1$	$F_0[x_0] = 1$	$F_1[x_0,x_1] = \frac{F_0[x_0]-F_0[x_1]}{x_0-x_1} = 1$	$F_2[x_0,x_1,x_2]$
$x_1 = 2$	$F_0[x_1] = 2$	$F_1[x_1,x_2] = \frac{F_0[x_1]-F_0[x_2]}{x_1-x_2} = -2$	$F_2[x_1,x_2,x_3]$
$x_2 = 3$	$F_0[x_2] = 0$	$F_1[x_2,x_3] = \frac{F_0[x_2]-F_0[x_3]}{x_2-x_3} = 1$	
$x_3 = 4$	$F_0[x_3] = 1$		

$$F_2[x_0,x_1,x_2] = \frac{F_1[x_0,x_1]-F_1[x_1,x_2]}{x_0-x_2} = -\frac{3}{2}$$

$$F_2[x_1,x_2,x_3] = \frac{F_1[x_1,x_2]-F_1[x_2,x_3]}{x_1-x_3} = \frac{3}{2}$$

$$F_3[x_0,x_1,x_2,x_3] = \frac{F_2[x_0,x_1,x_2]-F_2[x_1,x_2,x_3]}{x_0-x_3} = 1$$

$$P_3(x) = f(x_0) + (x-x_0)F_1[x_0,x_1] + (x-x_0)(x-x_1)F_2[x_0,x_1,x_2] +$$
$$+ (x-x_0)(x-x_1)(x-x_2)F_3[x_0,x_1,x_2,x_3] =$$
$$= 1 + (x-1) + (x-1)(x-2)\cdot\left(-\frac{3}{2}\right) + (x-1)(x-2)(x-3)$$

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Interpolare liniară

$x_0 < x_1 < \dots < x_n$

SPLINE: $S : [x_0, x_n] \to \mathbb{R}$; $S_i : [x_i, x_{i+1}] \to \mathbb{R}$

$$S(x) = \begin{cases} S_0(x), & x \in [x_0, x_1) \\ S_1(x), & x \in [x_1, x_2) \\ \dots\dots \\ S_i(x), & x \in [x_i, x_{i+1}) \\ \dots\dots \\ S_{n-1}(x), & x \in [x_{n-1}, x_n] \end{cases}$$

$S_i(x) = a_i x + b_i$, $i = 0 : n-1$

$a_i, b_i = ?$

Metoda 1

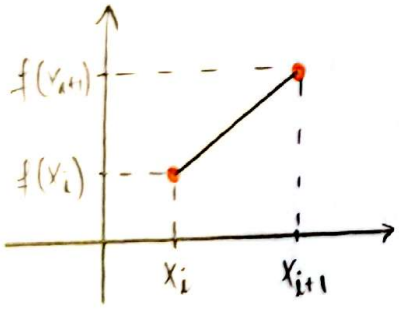
condiții de interpolare: $S_i(x_i) = f(x_i)$, $i = 0 : n-1$

$S_{n-1}(x_n) = f(x_n)$

condiții de racordare: $S_i(x_{i+1}) = S_{i+1}(x_{i+1})$; $i = 0 : n-2$

Metoda 2 (interpretare geometrică)

$S_i : [x_i, x_{i+1}] \to \mathbb{R}$, $S_i(x) = a_i \cdot x + b_i$



$$\frac{y - f(x_i)}{f(x_{i+1}) - f(x_i)} = \frac{x - x_i}{x_{i+1} - x_i}$$

$$y = f(x_i) + \frac{(x - x_i)[f(x_{i+1}) - f(x_i)]}{x_{i+1} - x_i}$$

$$y = f(x_i) + \frac{x f(x_{i+1}) - x f(x_i) - x_i f(x_{i+1}) + x_i f(x_i)}{x_{i+1} - x_i}$$

$$y = \underbrace{\frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}}_{a_i} x + \underbrace{\frac{f(x_i) x_{i+1} - f(x_{i+1}) x_i}{x_{i+1} - x_i}}_{b_i}$$

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Interpolare cubică de clasă C_1

$$s_i(x) = a_i + b_i(x-x_i) + c_i(x-x_i)^2 + d_i(x-x_i)^3, \quad i = 0:m-1$$

$s_i : [x_i, x_{i+1}] \to \mathbb{R}$

$a_i, b_i, c_i, d_i = ?$

- condiții de interpolare de tip Hermite:

$$s_i(x_i) = f(x_i), \quad i = 0:m-1$$

$$s_{m-1}(x_m) = f(x_m)$$

$$s_i'(x_i) = f'(x_i), \quad i = 0:m-1$$

$$s_{m-1}'(x_m) = f'(x_m)$$

- condiții de racordare:

$$s_i(x_{i+1}) = s_{i+1}(x_{i+1}), \quad i = 0:m-2$$

Spline-ul cubic de clasă C_1 în formă parametrică

$$s_i(t) = a_i + b_i h_i t + c_i h_i^2 t^2 + d_i h_i^3 t^3, \quad i = 0:m-1$$

unde $h_i = x_{i+1} - x_i$; h_i este lungimea $[x_i, x_{i+1}]$

$$t = \frac{x - x_i}{h_i}$$

Spline-ul cubic de clasă C_1 în formă parametrică și folosind polinoamele Bernstein de gradul 3

$$s_i(t) = a_i'(1-t)^3 + b_i' \cdot 3(1-t)^2 t + c_i' 3(1-t)t^2 + d_i' \cdot t^3, \quad i = 0:m-1$$

unde

$$a_i' = f(x_i)$$

$$b_i' = f(x_i) + \frac{h_i}{3} f'(x_i)$$

$$c_i' = f(x_{i+1}) - \frac{h_i}{3} f'(x_{i+1})$$

$$d_i' = f(x_{i+1})$$

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Exemplu

x	$1 = x_0$	$2 = x_1$	$4 = x_2$
$f(x)$	3	4	6
$f'(x)$	0	2	5

$S_0 : [1,2] \to \mathbb{R}$

$S_1 : [2,4] \to \mathbb{R}$

- **Metoda 1** : $S_i(x) = a_i + (x-x_i)b_i + (x-x_i)^2 c_i + (x-x_i)^3 d_i$, $i = 0:1$

$S_i'(x) = b_i + 2(x-x_i)c_i + 3(x-x_i)^2 d_i$, $i = 0:1$

cond de interpolare:

$S_0(1) = f(1) = 3$ $\quad S_0'(1) = 6$

$S_1(2) = f(2) = 4$ $\quad S_1'(2) = 2$

$S_1(4) = f(4) = 6$ $\quad S_1'(4) = 5$

condiții de racordare:

$S_0(2) = S_1(2)$

$S_0'(2) = S_1'(2)$

- **Metoda 2**

$S_i(t) = a_i'(1-t)^3 + 3b_i'(1-t)^2 t + 3c_i'(1-t)t^2 + d_i' t^3$; $i = 0:1$

$h_0 = x_1 - x_0 = 2-1 = 1$; $h_1 = x_2 - x_1 = 4-2 = 2$

$a_0' = 3$ $\quad d_0' = 4$ $\quad b_0' = 3 + \frac{1}{3}\cdot 0 = 3$ $\quad c_0' = 4 - \frac{1}{3}\cdot 2 = \frac{10}{3}$

$a_1' = 4$ $\quad d_1' = 6$ $\quad b_1' = 4 + \frac{2}{3}\cdot 2 = \frac{16}{3}$ $\quad c_1' = 6 - \frac{2}{3}\cdot 5 = \frac{8}{3}$

$\Rightarrow S_0(t) = 3(1-t)^3 + 9(1-t)^2 \cdot t + 10(1-t)\cdot t^2 + 4t^3$

$S_0(t) \to S_0(x)$; se face schimbarea de variabilă $t = \frac{x-x_0}{h_0} = x-1$

$\Rightarrow S_0(x) = 3(2-x)^3 + 9(2-x)^2(x-1) + 10(2-x)(x-1)^2 + 4(x-1)^3$.

Similar, obținem $S_1(t)$ și $S_1(x)$

$S_1(t) = 4(1-t)^3 + 16(1-t)^2 t + 8(1-t)t^2 + 6t^3$

$S_1(t) \to S_1(x)$; se face schimbarea de variabilă $t = \frac{x-x_1}{h_1} = \frac{x-2}{2}$

$\Rightarrow S_1(x) = 4\left(1 - \frac{x-2}{2}\right)^3 + 16\left(1-\frac{x-2}{2}\right)^2 \cdot \frac{x-2}{2} + 8\left(1-\frac{x-2}{2}\right)\frac{x-2}{2} + 6\cdot\frac{(x-2)^3}{8}$

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Interpolare cubică de clasă C_2

$S_i(x) = a_i + (x-x_i)b_i + (x-x_i)^2 c_i + (x-x_i)^3 d_i$, $i = 0:m$

- condiții de interpolare

$S_i(x_i) = f(x_i)$, $i = 0:m-1$

$S_{m-1}(x_m) = f(x_m)$

- condiții de racordare

$S_i(x_{i+1}) = S_{i+1}(x_{i+1})$, $i = 0:m-2$

$S_i'(x_{i+1}) = S_{i+1}'(x_{i+1})$, $i = 0:m-2$

$S_i''(x_{i+1}) = S_{i+1}''(x_{i+1})$, $i = 0:m-2$

Spline cubic de clasă C_2 natural: $S_0''(x_0) = S_{m-1}''(x_m) = 0$

Spline cubic de clasă C_2 tensionat: $S_0'(x_0) = f'(x_0)$

$S_{m-1}'(x_m) = f'(x_m)$

Exemplu

x	-1	0	1	2
$f(x)$	-2	-1	2	4
$f'(x)$	3			2

$S_0 : [-1, 0] \to \mathbb{R}$

$S_1 : [0, 1] \to \mathbb{R}$

$S_2 : [1, 2] \to \mathbb{R}$

$S_i(x) = a_i + (x-x_i)b_i + (x-x_i)^2 c_i + (x-x_i)^3 d_i$, $i = 0:2$

condiții de interpolare: $S_0(-1) = -2$; $S_1(0) = -1$; $S_2(1) = 2$

$S_2(2) = 4$

condiții de racordare: $S_0(0) = S_1(0)$, $S_1(1) = S_2(1)$, $S_0'(0) = S_1'(0)$

$S_1'(1) = S_2'(1)$, $S_0''(0) = S_1''(0)$, $S_1''(1) = S_2''(1)$

C_2 natural: $S_0''(-1) = S_2''(2) = 0$

C_2 tensionat: $S_0'(-1) = 3$; $S_2'(2) = 2$

$S_i'(x) = b_i + 2c_i(x-x_i) + 3d_i(x-x_i)^2$, $i = 0:2$

$S_i''(x) = 2c_i + 6d_i(x-x_i)$, $i = 0:2$

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Curbe Bezier

Fie $P_0(x_0, y_0)$, $P_1(x_1, y_1)$, ..., $P_m(x_m, y_m)$

P_0, P_m se numesc puncte de interpolare

$P_1, \dots, P_{m-1}$ se numesc puncte de control.

- curba Bezier de ordinul n: $B_n(t) = \sum_{i=0}^{n} P_i \cdot B_{i,n}(t)$

$t \in [0,1]$

$B_{i,n}(t)$ → polinom Bernsteim de gradul n.

$n=1$: Curba Bezier liniară (2 pct de intrare: P_0, P_1)

$$B_1(t) = P_0 \cdot B_{0,1}(t) + P_1 \cdot B_{1,1}(t) = P_0(1-t) + P_1 \cdot t, \quad t \in [0,1]$$

interpretare geometrică:

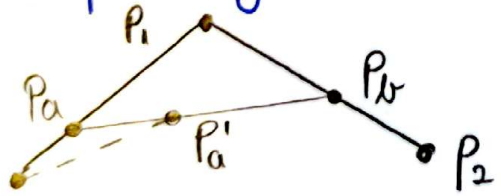
P_0 • ..... • P_1

$t \in [0,1]$

$n=2$: Curba Bezier pătratică (3 pct de intrare: P_0, P_1, P_2)

$$B_2(t) = P_0 \cdot B_{0,2}(t) + P_1 \cdot B_{1,2}(t) + P_2 \cdot B_{2,2}(t) =$$
$$= P_0(1-t)^2 + P_1 \cdot 2t(1-t) + P_2 \cdot t^2$$

interpretare geometrică:



$P_0 \to P_a = (1-t)P_0 + tP_1 \to P_a' = B_2(t) = (1-t)P_a + tP_b$

$P_1 \to P_b = (1-t)P_1 + tP_2 \to$ unde t fixat în $[0,1]$

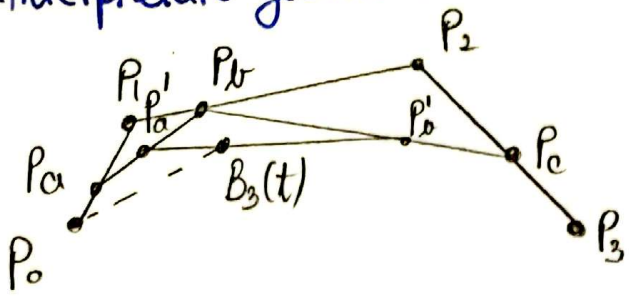
$P_2 \to$

$n=3$: Curba Bezier cubică (4 pct de intrare: P_0, P_1, P_2, P_3)

$$B_3(t) = P_0(1-t)^3 + 3P_1(1-t)^2 t + 3P_2(1-t)t^2 + P_3 t^3, \quad t \in [0,1]$$

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interpretare geometrică:



$P_0 \rightarrow P_a = (1-t)P_0 + tP_1 \rightarrow P_a' = (1-t)P_a + tP_b \rightarrow B_3(t) = (1-t)P_a' + tP_b'$,

$P_1 \rightarrow P_b = (1-t)P_1 + tP_2 \rightarrow P_b' = (1-t)P_b + tP_c$

$P_2 \rightarrow P_c = (1-t)P_2 + tP_3$

P_3

t fixat în $[0,1]$

Algoritmul de Casteljau

Date de intrare: $P_1(x_1, y_1)$; ... ; $P_n(x_n, y_n)$; $t \in [0,1]$

```
for j = 2 : n
    for i = 1 : n-j+1
        P_i^(j) = (1-t) P_i^(j-1) + t P_(i+1)^(j-1)
return P_1^(n)
```


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Aproximare numerică CURS 17

Fie spațiul prehilbertian (F, μ), unde F spațiu vectorial și μ produs scalar.

$\mu : F \times F \to \mathbb{R}$

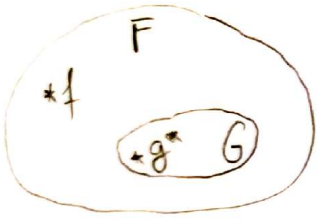
$\mu(f_1, f_2) \overset{not}{=} \langle f_1, f_2 \rangle$

ex: $F = \mathbb{R}^m \Rightarrow \langle x_1, x_2 \rangle = \sum_{i=1}^{m} x_i y_i \; ; \; x, y \in \mathbb{R}^m$

Proprietăți produs scalar

1) $\langle f_1, f_2 \rangle = \langle f_2, f_1 \rangle$
2) $\langle f_1 + f_2, f_3 \rangle = \langle f_1, f_3 \rangle + \langle f_2, f_3 \rangle$
3) $\langle f_1, f_2 \rangle \geq 0$
4) $\langle c \cdot f_1, f_2 \rangle = c \cdot \langle f_1, f_2 \rangle$

Fie G un subspațiu de dimensiune finită al lui F:



Cel mai bun aproximant în sensul celor mai mici pătrate pt elementul $f \in F$ în subspațiul G este $g^* \in G$ a.î.

$$\|f - g^*\|_2 = \min_{g \in G} \{\|f - g\|_2\}$$

Condiția necesară și suficientă pt ca g^* să fie cel mai bun aproximant în sensul celor mai mici pătrate pt elem $f \in F$ este:

$$\langle f - g^*, g \rangle = 0.$$

Fie $\{\mu_0, \mu_1, \dots, \mu_m\}$ bază în G. $g^* = \sum_{k=0}^{m} c_k^* \mu_k$

g element oarecare din G, $g = \sum_{j=0}^{m} c_j \cdot \mu_j \; ; \; g \neq g^*$

$$\langle f - g^*, g \rangle = 0 \Leftrightarrow \langle f - g^*, \sum_{j=0}^{m} c_j \cdot \mu_j \rangle = 0 \Leftrightarrow$$

$$\Leftrightarrow \langle \sum_{j=0}^{m} c_j \cdot \mu_j, f - g^* \rangle = 0 \Leftrightarrow \sum_{j=0}^{m} c_j \langle \mu_j, f - g^* \rangle = 0 \Leftrightarrow$$

$$\Leftrightarrow \langle \mu_j, f - g^* \rangle = 0 \Leftrightarrow \langle \mu_j, f - \sum_{k=0}^{m} c_k^* \mu_k \rangle = 0 \Leftrightarrow$$

$$\Leftrightarrow \langle \mu_j, \sum_{k=0}^{m} c_k^* \mu_k \rangle = \langle \mu_j, f \rangle, \; \forall j = 0 : m \Leftrightarrow$$

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$$\Rightarrow \sum_{k=0}^{m} \langle u_j, c_k^* u_k \rangle = \langle u_j, f \rangle \quad (2)$$

$$(\Leftrightarrow) \sum_{k=0}^{m} c_k^* \langle u_j, u_k \rangle = \langle u_j, f \rangle, \forall j = 0:m$$

$$\begin{bmatrix} \langle u_0,u_0\rangle & \langle u_0,u_1\rangle & \dots & \langle u_0,u_m\rangle \\ \dots & \dots & \dots & \dots \\ \langle u_m,u_0\rangle & \langle u_m,u_1\rangle & \dots & \langle u_m,u_m\rangle \end{bmatrix} \begin{bmatrix} c_0^* \\ \dots \\ c_m^* \end{bmatrix} = \begin{bmatrix} \langle u_0,f\rangle \\ \dots \\ \langle u_m,f\rangle \end{bmatrix}$$

simetrică

Bază ortogonală: $\langle u_i, u_j \rangle = 0$, $i \neq j$ $\Rightarrow$

$\Rightarrow$ matricea sist este diagonală

$$c_k^* = \frac{\langle u_k, f \rangle}{\langle u_k, u_k \rangle}, \quad k = 0:m$$

Bază ortonormată: $\langle u_i, u_j \rangle = 0$, $i \neq j$

$\langle u_i, u_i \rangle = 1$

$$c_k^* = \langle u_k, f \rangle, \quad k = 0:m$$

Aproximare continuă în sensul CMMP

- produs scalar: $\langle f, g \rangle \overset{not}{=} \int_a^b f(x) g(x) w(x)\, dx$ ↳ fct pondere

$f, g \in C([a,b])$

- bază polinomială: $\{1, x, x^2, \dots, x^m\}$

$$g^*(x) = c_m^* x^m + c_{m-1}^* x^{m-1} + \dots + c_1^* x + c_0^*$$

Exemplu: Calculați polinomul de aproximare în sensul c.m.m.p. de gradul 2 care aproximează funcția

$f(x) = |x|_1$, $x \in [-1,1]$, $w(x) = 1$.

$g_2^*(x) = c_2^* x^2 + c_1^* x + c_0^*$; $u_0(x) = 1$; $u_1(x) = x$; $u_2(x) = x^2$

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$$\begin{bmatrix} \langle u_0,u_0\rangle & \langle u_0,u_1\rangle & \langle u_0,u_2\rangle \\ \langle u_1,u_0\rangle & \langle u_1,u_1\rangle & \langle u_1,u_2\rangle \\ \langle u_2,u_0\rangle & \langle u_2,u_1\rangle & \langle u_2,u_2\rangle \end{bmatrix} \begin{bmatrix} c_0^* \\ c_1^* \\ c_2^* \end{bmatrix} = \begin{bmatrix} \langle f,u_0\rangle \\ \langle f,u_1\rangle \\ \langle f,u_2\rangle \end{bmatrix}$$

$$\langle u_0,u_0\rangle = \int_{-1}^{1} 1\cdot 1\cdot 1\,dx \qquad \langle u_0,u_1\rangle = \langle u_1,u_0\rangle = \int_{-1}^{1} 1\cdot x\cdot 1\,dx$$

$$\langle u_1,u_1\rangle = \int_{-1}^{1} x\cdot x\cdot 1\,dx \qquad \langle u_1,u_2\rangle = \langle u_2,u_1\rangle = \int_{-1}^{1} x\cdot x^2\cdot 1\,dx$$

$$\langle u_2,u_2\rangle = \int_{-1}^{1} x^2\cdot x^2\cdot 1\,dx \qquad \langle u_0,u_2\rangle = \langle u_2,u_0\rangle = \int_{-1}^{1} 1\cdot x^2\cdot 1\,dx$$

$$\langle f,u_0\rangle = \int_{-1}^{1} 1\cdot |x|\cdot 1\,dx \qquad \langle f,u_1\rangle = \int_{-1}^{1} x\,|x|\cdot 1\,dx$$

$$\langle f,u_2\rangle = \int_{-1}^{1} x^2\cdot |x|\cdot 1\,dx.$$

Rezolvând SEL se află c_1^*, c_0^*, c_2^*

Bază trigonometrică

$$\left\{\frac{1}{\sqrt{2}}, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx\right\}$$

$$\frac{1}{\pi}\int_0^{2\pi} u_q(x)\cdot u_r(x)\,dx = \begin{cases} 0, & q \neq r \\ 1, & q = r \end{cases}$$

$$g^*(x) = \frac{1}{\sqrt{2}}a_0 + \sum_{p=1}^{n}\left(a_p\cos px + b_p \sin px\right)$$

$$a_0 = \langle u_0, f\rangle = \frac{1}{\pi}\cdot\frac{1}{\sqrt{2}}\int_0^{2\pi} f(x)\,dx$$

$$a_p = \langle \cos px, f\rangle = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos px\,dx,\ p=1:n$$

$$b_p = \langle \sin px, f\rangle = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin px\,dx,\ p=1:n$$

Baza Cebâsev

$$\left\{\frac{1}{\sqrt{2}}, T_1(x), \dots, T_n(x)\right\}$$

Polinomul Cebâsev de ordin n:

$$T_n(x) = \cos(n \arccos x),\ n \geq 0$$

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Notăm: $\theta = \arccos x$, $\theta \in [0, \pi]$

$T_n(x) = \cos(n\theta)$

$T_{n+1}(x) = \cos((n+1)\theta) = \cos n\theta \cos\theta - \sin n\theta \sin\theta$ (1)

$T_{n-1}(x) = \cos((n-1)\theta) = \cos n\theta \cos\theta + \sin n\theta \sin\theta$ (2)

$\xrightarrow{(1)+(2)}$ $T_{n+1}(x) + T_{n-1}(x) = 2\underbrace{\cos n\theta}_{T_n(x)}\underbrace{\cos\theta}_{x} = 2T_n(x)\cdot x$

$\Rightarrow$ $\boxed{T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x) \;;\; T_0(x) = 1 \;;\; T_1(x) = x}$

$$\int_{-1}^{1} \frac{T_q(x)\cdot T_r(x)}{\sqrt{1-x^2}}\,dx = \begin{cases} 0, & q \neq r \\ \frac{\pi}{2}, & q = r \neq 0 \\ \pi, & q = r = 0 \end{cases}$$

$$g^*(x) = \frac{1}{\sqrt{2}}a_0 + \sum_{p=1}^{m} a_p \cdot T_p(x)$$

$$a_0 = \left\langle \frac{1}{\sqrt{2}}, f \right\rangle = \frac{1}{\pi}\cdot\frac{1}{\sqrt{2}}\int_{-1}^{1}\frac{f(x)}{\sqrt{1-x^2}}\,dx$$

$$a_p = \langle T_p, f\rangle = \frac{2}{\pi}\int_{-1}^{1}\frac{T_p(x)f(x)}{\sqrt{1-x^2}}\,dx,\quad p = 1:m$$

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